RailSentinel-24

Control-power monitor for DC fast-charger cabinets

PRELIMINARY

RailSentinel-24 (RS-24) mounts on the DIN rail inside a DC fast-charger cabinet and watches the 24 V control supply that every relay, contactor and controller depends on. It is designed to record the transients that reset controllers before they can write a log, timestamp them against a battery-backed clock, store them on board and report them over Wi-Fi — independently of the charger's own controller. Target: the transient-capture hardware is in design and the firmware is in development. The primary (sensing) side and the secondary (communication) side are separated on the board by an isolation barrier.

0–62 V	±0.5 %	9–32 V	≥ 1,000
measuring range · 12 / 24 V DC control rails	rail voltage after calibration · target	operating input · DIN rail	events stored on board · target

Firmware is in development. Lines marked "Target" are design targets, confirmed by test before release. "Hardware in design" marks functions whose circuit is not drawn yet.

POWER INPUT

Supply	from the monitored rail	no separate supply
Rails	12 / 24 V DC control rails	nominal on the per-unit label
Operating input	9–32 V	12 V and 24 V rails. Target (hardware in design): on a 12 V rail the power-loss record is a summary record. A 48 V variant is not decided.
Protection	TVS clamp, reverse polarity	design target: 70 V / 1 ms

RAIL MEASUREMENT

Measuring range	0–62 V	no clamp below 62 V: overvoltage up to 62 V is recorded in full; above 62 V the reading saturates
Accuracy	±0.5 % after calibration	TARGET · FIRMWARE IN DEVELOPMENT
Ripple trending	rolling baseline, PSU-degradation indicator	TARGET · FIRMWARE IN DEVELOPMENT
Current	with the clamp module	PLANNED · V2

ISOLATION AND FUNCTION

Barrier	designed for reinforced insulation	sensing side vs. communication side
Function	monitoring only	RS-24 does not control, inhibit or switch any charger function
Self-supervision	The measuring controller watches the communication controller. If its heartbeat stops, the alarm contact opens.	TARGET · HARDWARE IN DESIGN

J3 alarm contact. J3 (OK-CONTACT) is a potential-free, normally-open photoMOS contact. Contact rating: 24 V DC, 2–30 mA; for higher loads use an interposing relay. J3 is a diagnostic contact, not a safety-chain input. *Target behaviour (design in progress):* closed while the unit is monitoring; opens at power-loss stage 2 (rail < 0.667 × nominal, e.g. 16 V on 24 V), on reset, and while the communication side is unpowered; closes again ≤ 2 s after recovery. Typical open time on a deep rail dip with recovery: the dip duration + ≈ 1.5 s (≈ 2 s for a 10–100 ms interruption). Recommended PLC input debounce: ≥ 3 s for an alarm, no debounce for logging.

TRANSIENT CAPTURE

Trigger	hardware comparator, µs class	TARGET · HARDWARE IN DESIGN
Alert threshold	adjustable	per site, from firmware
Peak amplitude	per event, 0–62 V; accuracy published after bench test	TARGET · HARDWARE IN DESIGN
Waveform	50 ms at 10 µs per event	TARGET · HARDWARE IN DESIGN
Timestamp	±2 ppm RTC (0 … +40 °C)	battery-backed, valid offline; wider range after the thermal test

Storage	≥ 1,000 events, the record survives a power loss	TARGET · HARDWARE IN DESIGN

ENVIRONMENT

Operating	under test	cabinet interior · range published after the thermal test
Storage	-20 … +85 °C	
Humidity	5–95 % RH	non-condensing · design target
Vibration / shock	tests to IEC 60068-2-6 / -27 planned; levels to be defined	

CONNECTIVITY

Wi-Fi	2.4 GHz	all variants, via the site network
Ethernet	1 port, COM-ETH module	PLANNED · V2
Telemetry	MQTT, alert within 5 s of an event	TARGET · FIRMWARE IN DEVELOPMENT
Site integration	Modbus TCP over Wi-Fi · MQTT to a local broker, all variants	TARGET · FIRMWARE IN DEVELOPMENT
Updates	signed OTA firmware	TARGET · FIRMWARE IN DEVELOPMENT

VARIANTS

RS-24-S · SLIM
Slim
Narrow DIN module. Wi-Fi. For small CPOs.

RS-24-S · COUPLER
Standard
DIN coupler. Wi-Fi. Expansion slots for Ethernet (COM-ETH) and the clamp module for rail current PLANNED · V2

RS-24-M
OEM module
Bare host board 99 × 69 mm for factory integration. Modbus TCP · MQTT · signed OTA updates: target, firmware in development.

PLUG-IN MODULES PLANNED · V2

EXP-A (sensing side)	clamp current	
EXP-B (communication side)	RS-485 / Modbus · SD logger · DIP config	one option at a time
COM-ETH (inside the Standard coupler)	Ethernet, 1 port	
SENSE	temperature · sound · vibration	right-side module
Second COM	a second RS-485 line	right-side module

ON THE ROADMAP · NO COMMITMENT

NFC tap-to-read at the cabinet	BACKLOG	Tell us what you need: hello@morgen-tech.com, subject "RS-24 roadmap vote". Your votes decide what we build first.
Cellular field variant	BACKLOG	
Hold-up power	BACKLOG	

STATUS AND PILOTS

Product in development. No CE marking yet; not commercially available. Not a certified safety component today. Pilot units from Q2 2027, on loan under a test agreement.

RailSentinel-24 and RS-24 are product names of Morgen Technologie UG (haftungsbeschränkt). Specifications may change without notice.